Prague University of Economics and Business
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SOCIAL SPENDING AND ITS EFFECT ON ECONOMIC ACTIVITY ACROSS EUROPE

BACHELOR THESIS

Study program: Data Analytics

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Abstrakt

Tato bakalářská práce zkoumá vztah mezi vládními výdaji na sociální zabezpečení a ekonomickou výkonností, která je vyjádřena skrze HDP na obyvatele, napříč evropskými zeměmi v období let 2002 až 2022. Pomocí dat z Eurostatu a panelové regresní analýzy s diferencí, studie odhaluje silný, statisticky významný pozitivní vztah – každé euro sociálních výdajů na obyvatele je spojeno s růstem HDP přibližně o 2.94 €. Před vyloučením či ošetřením extrémních případů (Irsko, Norsko) efekt dále roste na 3,42 €. Výsledky vycházejí z keynesiánské ekonomické teorie a jsou podpořeny existující odbornou literaturou, přičemž naznačují, že sociální výdaje mohou významně přispívat k ekonomickému růstu v institucionálně stabilních zemích. Práce rovněž upozorňuje na určitá omezení – využití ročních dat, vyloučení dynamických modelů a rozlišení jednotlivých kategorií sociálních výdajů, a také nabízí doporučení dalšího výzkumu.

Klíčová slova

Sociální výdaje, Hospodářský růst, Panelová regrese, Keynesiánský multiplikátor

Abstract

This thesis investigates the relationship between government social spending and economic performance, measured by GDP per capita, across European countries from 2002 to 2022. Using Eurostat data and first differences panel regression models, the study finds a strong, statistically significant positive relationship, with each euro of per capita social spending associated with a GDP increase of approximately 2.94 €. Before excluding or accounting for outliers (Ireland and Norway), the effect was even higher at 3.42 €. Grounded in Keynesian economic theory, and supported by existing literature, the findings suggest that social spending contributes meaningfully to economic growth in institutionally stable contexts. The study highlights limitations including the use of annual data, the exclusion of dynamic models, and the absence of disaggregated spending categories, offering clear directions for future research.

Keywords

Social spending, Economic growth, Panel regression, Keynesian multiplier

Contents

Introduction

The relationship between government social spending and economic growth remains one of the most enduring and contested issues in public economics. Rooted in the ideological divide between Keynesian and neoclassical schools of thought, the debate continues to shape fiscal policy discussions around the world. On one side, the Keynesian perspective argues that public expenditure, especially on social welfare programs, can stimulate aggregate demand, reduce economic volatility, and foster long-term development (Musgrave, 1959). On the other, critics following Hayekian or neoclassical frameworks caution that excessive public spending may crowd out private investment, distort incentives, or reduce fiscal sustainability.

Empirical studies on this subject have produced a broad range of findings. Some have demonstrated positive effects of social spending in high-income countries with strong institutions, where well-managed public transfers support human capital development and economic resilience (Castles & Obinger, 2008; Cooray & Nam, 2024). Others, however, emphasize inefficiencies and diminishing returns, particularly in countries where public sector effectiveness is limited or where spending is poorly targeted (Okafor, 2010). Recent research has also pointed to the importance of institutional quality as a moderating factor in determining whether social expenditure contributes to or hinders economic growth. As such, the relationship is complex, conditional, and highly dependent on national context and governance capacity.

This thesis contributes to this ongoing debate by examining the relationship between government social spending and gross domestic product (GDP) per capita across a panel of European countries over a twenty-year period (2002–2022). Unlike many existing studies that focus on short timeframes or heterogeneous global samples, this research narrows its scope to the European context, where governance structures, data comparability, and fiscal institutions are relatively stable. This controlled environment offers an opportunity to isolate the economic impact of social spending more clearly, minimizing the confounding effects of political instability or macroeconomic volatility found in broader global analyses.

Using data from Eurostat and econometric techniques including the fixed effects (FE) model, this thesis aims to determine whether variations in per capita social expenditure are associated with measurable changes in economic performance. Importantly, the study also acknowledges the conceptual challenge that social spending is itself a component of GDP (via the government expenditure term in the GDP formula), and therefore interprets correlations with caution. Additionally, this thesis omits disaggregation of social spending categories due to scope limitations but lays the groundwork for further research in this direction.

By offering a long-run, regionally focused, and methodologically transparent analysis, this thesis provides new insights into the fiscal policy debate within Europe. It also contributes to the literature by highlighting both the economic magnitude of social spending effects and the methodological considerations - such as data selection, treatment of outliers, and model specification - that influence empirical findings. In doing so, it seeks to inform both academic inquiry and practical policy formulation, particularly in an era where governments continue to grapple with balancing welfare commitments and financial deficits.

Motivation

The relationship between government social spending and economic growth has long been a central theme in both academic inquiry and political discourse. Contemporary debates - often simplified in popular media, such as the humorous but ideologically loaded YouTube "rap battle" between Friedrich Hayek and John Maynard Keynes - highlight opposing views on the role of government in the economy and underscoring their continued relevance in public discourse. On one hand, proponents of Hayekian thought argue that excessive state intervention through welfare and entitlement programs distorts markets and hampers growth. On the other hand, the Keynesian perspective emphasizes the stabilizing and growth-enhancing effects of targeted public expenditure, especially during economic downturns. This ideological clash is mirrored in political debates over fiscal responsibility, redistribution, and the size of the welfare state, making the topic not only theoretically significant but also highly relevant to ongoing policy discussions.

This thesis seeks to contribute to this debate by empirically analyzing the relationship between social spending and economic performance across European countries over a twenty-year period. At its core, the question examines whether higher levels of public social expenditure correlate positively or negatively with GDP per capita. A positive relationship may lend empirical support to more interventionist, left-leaning fiscal policies, while a negative or neutral relationship could validate arguments in favor of government reduction often championed by the political right. While the analysis avoids explicit political claims, its findings can help clarify whether broad trends in social expenditure align with economic growth trajectories in relatively stable institutional contexts. In doing so, it bridges economic theory with political relevance, offering insights for scholars and policymakers alike.

1. Literature Review

1.1 Literary Research Introduction

In this literature research theoretical and empirical foundation will be provided for analyzing the relationship between social spending and economic activity, measured primarily by GDP per capita. As this paper aims to evaluate how government social spending impacts economic outcomes across European countries over a twenty-year period, it is essential to first understand the rationale for using GDP per capita as a dependent variable, the definitions and categories of social spending, and existing findings on their interdependence. Additionally, social spending will be examined in terms of different types, categories, and its relation to economic activity as well as its limitations in its use. Lastly, we will be looking at data choice for the methodology part as well as the key part of this thesis, which is what econometrics models will be utilized and why. This background informs both the model choice and data selection for the empirical part of the thesis.

1.2 Economic Activity

Let us delve into how economic activity is measured, what issues there are with its measurement and what options there are for this thesis in terms of the time span we have chosen and the selection of countries we are going to be examining. Before selecting a suitable indicator of economic activity for this thesis, it is important to understand the landscape of available measures and their respective advantages and limitations. Economic activity can be captured in various ways, ranging from traditional income measures like GDP to more multidimensional indices that account for social and environmental factors. Common alternatives to GDP include the human development index (HDI), which incorporates life expectancy, education, and income; the Genuine Progress Indicator (GPI), which adjusts GDP for environmental degradation and social well-being (Van den Bergh & Antal, 2014); and gross national income (GNI), which reflects the total income earned by a country's residents, including income from abroad (Costanza et al., 2009). These alternatives offer a more nuanced understanding of development but present challenges in terms of data availability, comparability, and temporal coverage, especially when analyzing a large sample of countries over two decades.

Unfortunately, these more nuanced indicators are rarely as readily available as traditional GDP. The World Bank and Eurostat acknowledge that while composite indicators like HDI or GPI offer broader insights, they often lack annual, harmonized data across a wide set of countries—particularly for emerging economies (Eurostat, 2023; World Bank, 2022). Moreover, the methodological complexity and data requirements for calculating these indicators make them less suitable for econometric modeling in a multi-country panel setting, this is further expanded upon by (Bleys, 2012), who looks at the alternatives to GDP per capita in terms of informing governmental policy decisions. All of this would suggest that it is preferable to utilize these alternatives in this research, however, that omits quite a few advantages it poses in comparison to its alternatives.

These advantages are mostly based on its availability, consistency in measurement and its regularity in terms of country differences. It reflects the average economic output per person and allows for straightforward cross-country comparisons (Barro & Sala-i-Martin, 2004). Although it does not account for income distribution, informal economies, or ecological sustainability, its strengths lie in availability, comparability, and integration into economic modeling frameworks (Stiglitz et al., 2009). Hence, given the scope of this thesis, which lies in analyzing social spending across European countries from 2002 to 2022 it is the more feasible economic activity indicator. Now that we have established why it is appropriate for use in terms of this paper let us briefly examine what GDP per capita establishes.

GDP per capita remains one of the most widely used indicators of economic performance and living standards. It reflects the total economic output of a country divided by its population, capturing average income and productivity levels (Barro & Sala-i-Martin, 2004). The standard GDP identity, defined as:

$$\mathrm{Y = C + I + G + (X - M)} \ , \tag{1.1}$$

where Y (output) equals C (consumption), I (investment), G (government expenditure) and X (exports) lowered by M (imports).

1.3 Social Spending

Now that we have examined how to measure economic activity in this thesis and the literature attached to it let us delve into social spending's scope and classification. When talking about social spending in this thesis we will be strictly referring to government social spending, which comprises expenditures aimed at improving individual and societal welfare, including support in areas such as health, pensions, unemployment, housing, and family assistance. According to the (OECD, 2019), social expenditure can be categorized into two main forms: direct and indirect. Direct social spending refers to government-provided cash transfers (e.g., pensions, unemployment benefits) and in-kind benefits (e.g., public education or healthcare). Indirect social spending includes tax breaks with social purposes (TBSPs), such as deductions for dependent children (Adema et al., 2011).

However, while distinctions between direct (e.g., cash benefits, in-kind services) and indirect (e.g., tax breaks with social purposes) social spending are conceptually important, this thesis does not differentiate between them. This decision is based on the limited scope of the research and the focus on capturing broader trends in government social expenditure across countries and time. Additionally, detailed disaggregation of these categories is often inconsistently reported or unavailable across the full time span and country sample used. An example of these inconsistencies would be the United Kingdom where according to the Financial Times (2024) only 10% of the local authorities submitted reliable data in terms of property assets, pension liabilities etc., which led to the National Audit Office refusing to sign off on public sector accounts. As such, total public social spending is used as a composite measure, drawing from consistent datasets such as Eurostat and the Organisation for Economic Co-operation and Development (OECD) Social Expenditure Database (SOCX), which allow for consistent, cross-national analysis over a twenty-year period.

1.4 Relationship between GDP per capita and Social Spending

The relationship between economic performance and social spending has long been a topic of debate in public finance literature. On one hand, Wagner's Law claims that as a country's economy grows, the demand for social services rises, naturally leading to increased government spending (Peacock & Wiseman, 1961). On the other hand, Keynesian economics argues that public expenditure, including social welfare programs, can actively stimulate economic activity, particularly during periods of economic downturn (Musgrave, 1959).

Empirical findings on this relationship remain mixed. In higher-income countries, increased social spending is often associated with GDP growth, largely due to its role in enhancing human capital, reducing poverty, and stabilizing consumption (Castles & Obinger, 2008). Conversely, in lower-income countries or those with governance challenges, the effect may be weaker or even negative due to misallocation, inefficiencies, or a lack of institutional capacity (Okafor, 2010). The effect of Wagner's law seems limited to developing countries, as in both extremes (developed/rich and underdeveloped countries) the law does not hold (Magazzino, 2010).

It is important to acknowledge a conceptual limitation in analyzing this relationship: government social spending is itself a component of GDP, specifically under the expenditure approach as part of government consumption. This built-in overlap means that any correlation between the two variables may, to some extent, be mechanical. A more refined analysis would ideally subtract social spending from GDP to isolate its independent effects. However, due to the limited scope and data constraints of this thesis, such an adjustment will not be made. Instead, the relationship is interpreted with this limitation in mind, focusing on broad trends rather than causal inference.

Another study which looked at this relationship was done by Cooray and Nam (2024), which shows that public social spending can support economic growth, but only when paired with strong governance. Their global study highlights the importance of institutional quality in realizing the benefits of social expenditure. This thesis, however, builds on their work by focusing on European countries over a twenty-year period, offering a more region-specific analysis within relatively stable institutional contexts.

1.5 Data Justification

Eurostat is the primary data source for this thesis due to its consistency, methodological transparency, and alignment with the COFOG (Classification of the Functions of Government) framework. It provides harmonized and regularly updated data across nearly all European countries, making it particularly suitable for long-term panel analysis. While other sources such as the OECD Social Expenditure Database also offer detailed information on social spending - including distinctions between direct and indirect expenditure - they do not consistently cover all European countries and often report data for only selected years (e.g., every second year). In contrast, Eurostat provides more complete annual coverage over the 2002–2022 period, which is essential for the econometric models used in this thesis.

1.6 Econometric Models Used in the Literature

To analyze the relationship between social spending and economic activity, this thesis employs several econometric techniques commonly used in panel data analysis. Initially, a first-difference estimator is utilized to eliminate time-invariant country-specific effects, addressing the issue of unobserved heterogeneity and reducing potential biases from omitted variables (Wooldridge, 2010). To capture temporal trends and mitigate further omitted variable bias, a time trend is incorporated by including dummy variables for each year, excluding the initial year to prevent perfect multicollinearity with the intercept. Additionally, standard errors are clustered at the country level to account for within-country correlations over time, enhancing the robustness of statistical inferences (Cameron & Miller, 2015).

Subsequently, the analysis employs robust linear modeling (RLM) to further ensure robustness against potential outliers and leverage points that could disproportionately influence the estimation outcomes (Huber & Ronchetti, 2009). The choice of RLM is particularly important, given that panel data analyses often involve heterogeneous data points across countries and years, potentially skewing standard regression results.

Several empirical studies have applied these models to investigate similar relationships. For example, Cooray and Nam (2024) use FE models to examine how government effectiveness conditions the impact of social spending on growth. Castles and Obinger (2008) employ random effect (RE) models to assess the impact of welfare state regimes on economic outcomes in OECD countries. Similarly, Lindert (1996) underscored how robust econometric approaches are essential for accurately assessing the growth effects of welfare policies. Arellano-Bond dynamic panel models are also commonly used in macroeconomic research where lagged dependent variables play a role (Roodman, 2009), though these are more complex and data-intensive.

2. Methodology

2.1 Research Design and Objective

This thesis employs a quantitative, panel data methodology to investigate the relationship between government social spending and economic activity, measured by GDP per capita, across European countries over the period 2002–2022. The primary objective is to determine whether variations in government social expenditure correspond with observable changes in economic output on a per capita basis, while accounting for both temporal and cross-sectional heterogeneity.

Panel data techniques are particularly well-suited to this research objective as they enable control for unobserved country-specific characteristics that may bias the estimates in simple cross-sectional or time-series models. The use of multiple countries over a twenty-year span allows for a richer analysis that captures both within-country dynamics and between-country differences in fiscal policy and economic performance.

2.2 Data Sources and Justification

The empirical analysis relies primarily on data from Eurostat, which provides annual indicators for European Union member states and several associated countries. Eurostat's adherence to the Classification of the Functions of Government framework ensures comparability across national statistics offices and alignment with international reporting standards.

Alternative data sources such as the OECD Social Expenditure Database were considered. The SOCX database offers greater detail in distinguishing between direct and indirect forms of social spending, including tax breaks with social purposes. However, its use is constrained by uneven country coverage and irregular time intervals (often providing data only for odd years or limited subperiods), which would substantially reduce the completeness and comparability of the panel dataset. Eurostat, by contrast, provides near-complete annual data across most European countries, making it more appropriate for the continuous and longitudinal scope of this study.

Table 2.1

Example of input data – Social spending in Euro per capita (only a sample of countries in limited years included)

Country	2002	2003	2004	2005	2006	2007	2008
Belgium	6657.37	7031.99	7374.73	7616.53	7838.28	8086.34	8652.40
Czechia	1534.63	1553.67	1619.92	1862.53	2054.25	2281.50	2685.51
Denmark	9910.63	10488.48	10877.44	11246.57	11446.73	12582.43	12912.72
Germany	7568.16	7710.78	7691.12	7740.41	7736.33	7866.12	8135.96
Estonia	703.10	786.10	910.67	1021.15	1189.31	1433.94	1792.23
Ireland	5439.49	5847.73	6377.58	6698.36	7097.52	7566.95	8197.90
Greece	2695.46	2948.37	3244.13	3610.42	3981.82	4387.71	4888.02
Spain	3440.10	3675.57	3890.62	4180.90	4435.90	4727.57	5094.86
France	7241.82	7550.73	7854.66	8097.25	8331.76	8624.68	8894.70
Italy	5502.85	5742.52	5993.01	6197.28	6481.71	6669.84	6970.19
Cyprus	2490.99	2926.26	3073.54	3308.13	3549.62	3696.25	4049.33
Latvia	585.49	578.96	619.22	719.67	903.26	1064.69	1331.17

2.3 Variable Selection

Dependent Variable: GDP per capita (measured in nominal Euros), representing the average economic output per person.

Independent Variable: Social spending in Euros per capita, including all categories of social protection expenditure. As noted, this thesis does not distinguish between direct and indirect forms due to data availability constraints and the research scope.

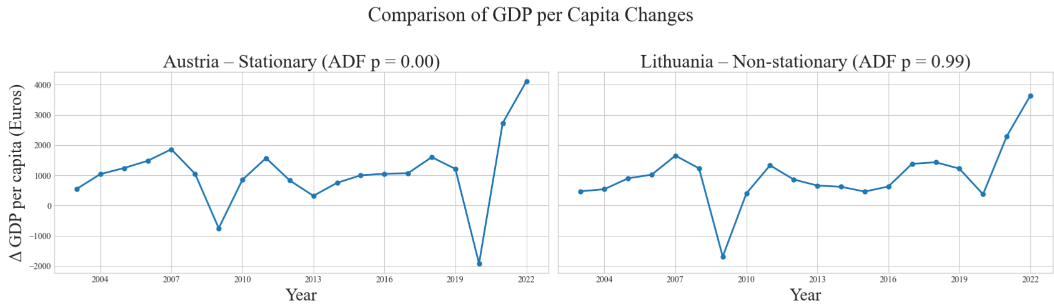
2.4 Econometric Modeling Strategy

To examine the effect of social spending on GDP per capita, the analysis proceeds in these stages:

1. The stationarity of the panel data was assessed using the Augmented Dickey-Fuller (ADF) test, which revealed that the data were largely non-stationary, indicating the presence of unit roots and raising concerns about potential spurious regression results (Wooldridge, 2010). Furthermore, the Engle-Granger test was employed to examine cointegration, with the majority of country-level tests suggesting a lack of cointegration (Engle & Granger, 1987).
2. To mitigate non-stationarity concerns, first differencing was applied to the data. Reapplying the ADF tests after differencing showed improved stationarity, with approximately 38% of countries meeting the formal criterion of stationarity.

Figure 2.1

Comparison of two extremes in ADF outputs



Closer inspection of countries that failed the ADF test even after differencing shows that they failed mostly because of recent positive shocks during the miraculous economic boom after the global pandemic. In Figure 2.1 we can see the two countries with the most extreme ADF results - with the main difference being how much of a economic hit the country received in 2020. For the remaining countries, visual inspection suggested that apparent non-stationarity was primarily due to temporary shocks rather than persistent trends. Thus, the differenced data would be treated as stationary for subsequent analysis.

3. Given these stationarity considerations and the fact that most countries were not cointegrated, a first difference estimator was employed to analyze the relationship between changes in social spending and GDP per capita. This estimator effectively controls for time-invariant, country-specific characteristics, addressing omitted variable bias due to unobserved heterogeneity (Wooldridge, 2010). Additionally, a time trend was included through dummy variables for each year (excluding the initial year) to control for temporal variations, and clustered standard errors were used to correct for within-country correlation across time (Cameron & Miller, 2015).
4. Recognizing the potential influence of outliers and leverage points inherent in panel datasets, diagnostic checks for outliers were performed. Subsequently, results from the first difference estimator were compared with outcomes obtained from robust linear modeling (Huber & Ronchetti, 2009). This comparative approach ensures robustness and reliability of the final econometric conclusions.

Although dynamic panel models such as the Arellano-Bond estimator (Roodman, 2009) are frequently used in growth regressions to account for time-lagged dependent variables and endogeneity, their application falls beyond the methodological scope of this thesis due to complexity, instrument validity requirements, and sample size limitations. Additionally, fixed effects and random effects models were not employed because differencing was required to achieve stationarity in the data (Wooldridge, 2010).

2.5 Software and Implementation

All analysis was conducted in Python using the following packages:

- `pandas` for data cleaning and time-series manipulation,
- `statsmodels` for econometric modeling and testing,
- `plotly` for interactive visualization and trend presentation.

Python was selected for its open-source nature, reproducibility, and strong integration of data science and statistical libraries, making it well-suited for applied economic analysis.

2.6 Missing Values

The dataset used for this thesis was largely complete, with annual observations available for the vast majority of countries over the full 2003–2023 period. However, three countries—Croatia, Bulgaria, and the United Kingdom—exhibited missing data that warranted exclusion.

Croatia and Bulgaria, which joined the European Union partway through the observed period, lacked consistent data for the early 2000s. Meanwhile, the United Kingdom had several missing entries following its departure from the EU, as it ceased standardized reporting to Eurostat.

Rather than risk introducing bias through imputation—particularly in a panel data context where temporal and cross-sectional consistency is critical—these countries were excluded from the final analysis. This decision ensured the reliability of the regression estimates while minimizing the impact of incomplete or inconsistent time series. Given the small number of omitted cases relative to the full panel, the overall integrity and representativeness of the dataset remain intact.

3. Results

This section presents the empirical findings from the first difference estimator (FDE) panel regression analysis examining the relationship between government social spending and economic performance, as measured by GDP per capita across 26 European countries from 2003 to 2022 (the original data was from 2002, but we lose a year by employing the FDE).

3.1 Model Selection and Specification

After assessing that a first difference estimator is a good choice for our analysis our initial model after adding time variables for each year recorded resulted in a social spending coefficient of 3.42. After inspecting the root mean square errors of the model two outliers were identified and two approaches were taken to assess and potentially lower their impact. After omitting the two outliers the coefficient lowered to about 2.95. When using a RLM with all countries included the model resulted in a very similar coefficient of 2.94.

To account for potential heteroskedasticity and serial correlation, clustered standard errors at the country level were used. The general specification of the FDE models is as follows:

$$\Delta\text{GDP}_{it} = \alpha + \beta \cdot \Delta\text{SocialSpending}_{it} + \sum_{t=2}^{T} \delta_t D_t + \Delta\varepsilon_{it} \ , \tag{3.1}$$

where ΔGDP_{it} denotes the change in GDP per capita for country i in year t, $\Delta\text{SocialSpending}_{it}$ represents the corresponding change in the change of (per capita) government social spending, and D_t are year-specific dummy variables (excluding the base year, as that is implicitly in the intercept) that capture macroeconomic shocks common across all countries. The parameter β estimates the average effect of changes in social spending on changes in GDP, while $\Delta\varepsilon_i t$ is the differenced error term.

3.2 Regression Output

After removing the three main outliers our PanelOLS estimation summary shows us that our model, based on 580 observations across 29 countries with a clustered covariance estimator, has a fairly high adjusted R-squared value of 0.515. After removing outliers we get an even higher of adjusted R-squared 0.624. This indicates that outliers do have a significant effect and thus a more robust approach should be used - so a robust linear model was used, with a similar coefficient as the FDE after removing outliers, indicating that it is a reasonable choice.

Table 3.1

Summary of Coefficients Across Models

Model	Estimate	Std. Error	P-value	95% CI
FDE	3.4190	0.730	0.000	[1.988, 4.850]
FDE (outliers omitted)	2.9461	0.671	0.000	[1.631, 4.261]
RLM	2.9402	0.081	0.000	[2.781, 3.099]

From these results we can see that our model is statistically significant with more than half of variability in GDP described by changes in social spending on average. This is partly because of the mechanical relationship of these variables, but presumably the effect extends beyond that. As our 95% confidence interval (Table 3.1) is comfortably above 0 we can conclude that the relationship between our two variables is positive according to the modeled estimate in any of our models used.

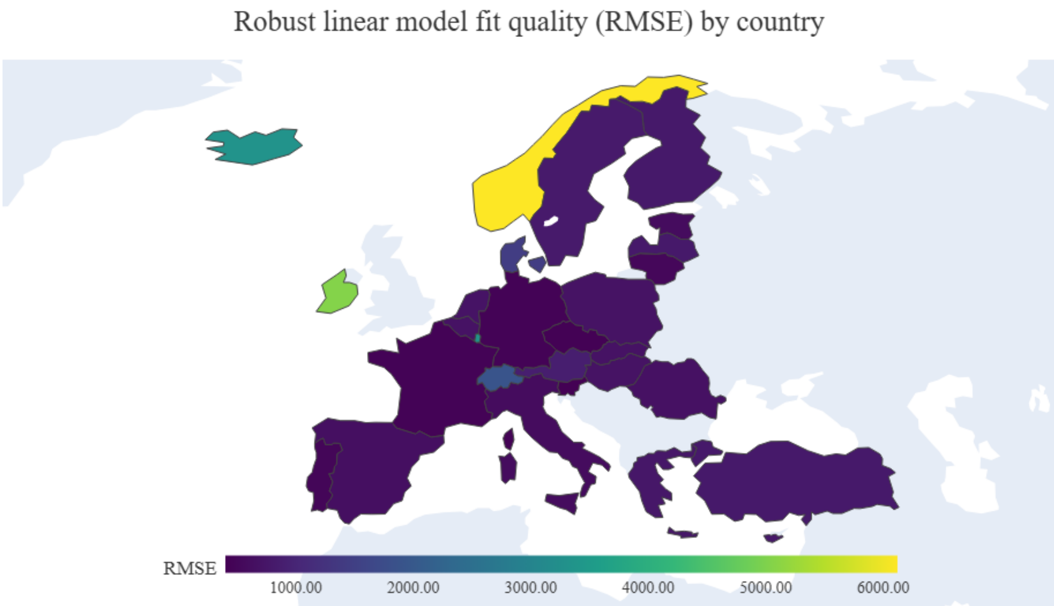


Figure 3.1

Choropleth Map of RLM Residual RMSE

Figure 3.1 demonstrates that there is variation in the RMSE residuals across European countries even while using a robust linear model. This indicates that the relationship between social spending and GDP per capita is not uniform throughout Europe. Some countries will have smaller residuals indicating a better fit to the model, while others will have larger residuals showing more of a deviation.

The results indicate a strong and statistically significant positive relationship between per capita government social spending and GDP per capita. Specifically, a one-euro increase in social spending per capita is associated, on average, with a 2.94 € increase in GDP per capita.

3.3 Model Fit and Interpretation

By adding time variables to account for macroeconomic shocks, individual years are added to the model, with significant (both statistically and in size) values. For instance, in the RL model we can see two extremely negative values in years 2009 (-2034 €) and 2020 (-2604 €), these can be easily explained by the global financial crisis in 2008 and later the COVID-19 effect in 2020. Conversely, after 2020, in both 2021 and 2022 we see values of over +2000 €, showing the economic euphoria after the global pandemic.

3.4 Outlier Analysis and Robustness Check

Inspection of model residuals revealed that Ireland and Norway were substantial outliers (as can be seen in Figure 3.2).

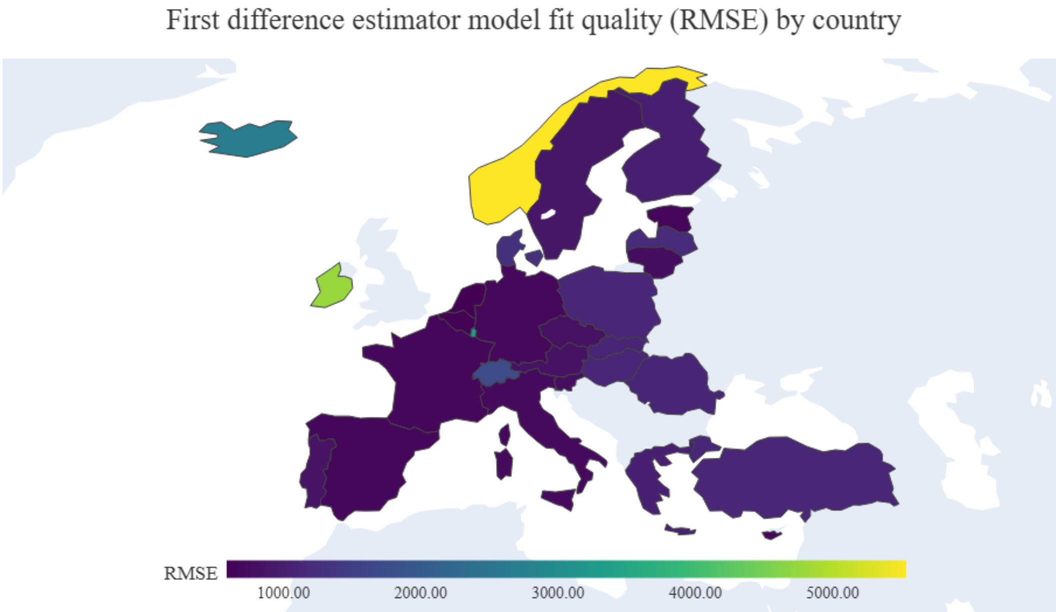


Figure 3.2

Choropleth Map of FD Model Residual RMSE (All Countries)

After removing these two countries from the dataset, the FD model was re-estimated. The coefficient on social spending remained statistically significant and positive, with a reduced magnitude of approximately 2.95 (compared to the original 3.42), reinforcing the robustness of the initial findings (even if lowering the values) and confirming the broader European trend in more typical fiscal environments.

In Figure 3.2, we can see that after removing the outliers, the maximum RMSE of the residuals per country lowered significantly from more than 5000 to just over 3000.

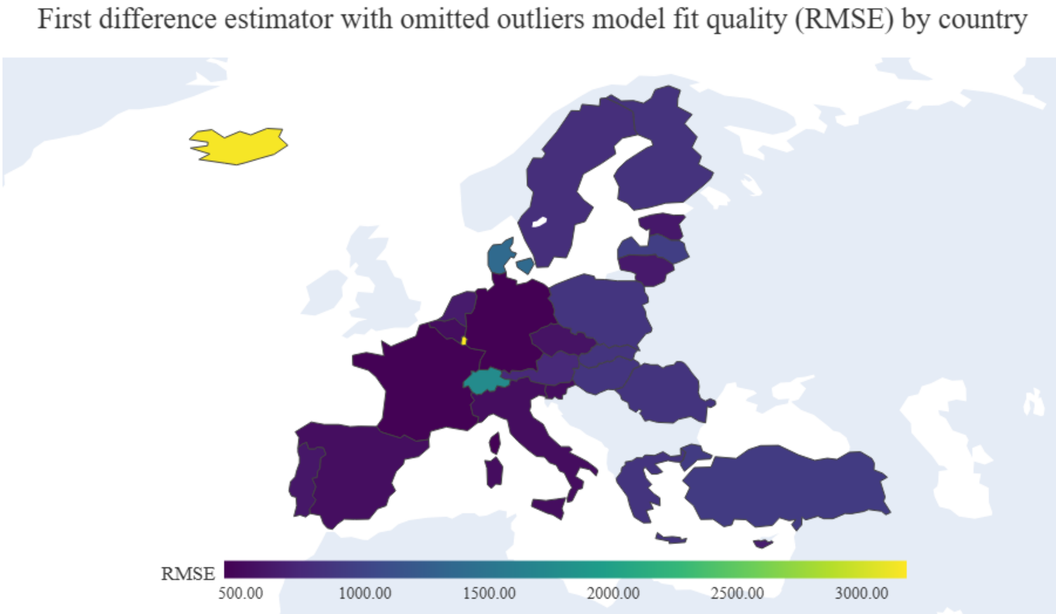


Figure 3.3

Choropleth Map of FD Model Residual RMSE (Outliers Removed)

From the significant difference after omitting outliers (and from the similarity of the RLM to the model with outliers omitted), the robust linear model was deemed to be more accurate and will be used in final findings.

3.5 Summary of Key Findings

- The positive relationship between social spending and GDP per capita is both statistically significant and economically meaningful.
- The first difference estimation reveals that approximately 2.94 € in GDP is associated with each additional euro of per capita social spending.

These findings align with Keynesian perspectives on public expenditure as a driver of economic activity, particularly in institutionally stable European contexts. The implications of these results are discussed further in the following section.

4. Analysis and Interpretation

4.1 Updated Keynesian Multiplier and Theoretical Context

After omitting the structural outliers Norway and Ireland from the first difference estimator (or use a RLM), the coefficient estimating the relationship between government social spending and GDP per capita decreases to 2.94. While lower than the previously estimated 3.42, the result remains both economically substantial and statistically robust, with a narrow confidence interval and high significance. This adjustment reinforces the stability of the relationship across the majority of European countries and indicates the strength of the Keynesian multiplier effect, even when outlier impact is reduced.

This result is best interpreted through the lens of the standard national income identity:

$$\mathrm{Y = C + I + G + (X - M)}\ , \tag{4.1}$$

where C represents consumption, I investment, G government expenditure, and $(X - M)$ net exports. Social spending, while accounted for under G (means that as we have to effectively lower the value by one to see the effects and not just the government investment itself to interpret the value correctly), has broader indirect effects on both C and I. Targeted government transfers and public services increase disposable income, particularly for lower-income households, who exhibit a higher marginal propensity to consume. In economic terms, each euro received by a lower-income household is more likely to be spent than saved—contributing directly to C. Unlike high-income households, whose additional income often goes into savings or non-productive financial instruments, lower-income groups funnel government transfers back into the real economy through consumption of goods and services.

Moreover, consistent social spending reduces income volatility and strengthens the social safety net, thereby creating a more predictable and secure environment for private sector investment. When basic needs such as healthcare, housing, and unemployment support are ensured, private actors face less uncertainty, which in turn stimulates I. This linkage - where social spending fosters both short-run demand and long-run investment stability - supports the theoretical framework behind Keynesian fiscal policy, particularly its counter-cyclical nature in smoothing out economic cycles and mitigating recessionary effects (Musgrave, 1959).

4.2 Ireland and Norway Outlier Analysis

The position of these two countries is grounded in their unique macroeconomic contexts, which distort the assumptions underlying the econometric model. Norway's GDP is heavily influenced by its sovereign wealth fund, which invests oil revenues globally and decouples economic performance from domestic consumption or public services. Ireland's data is distorted by multinational profit-shifting in so-called "Leprechaun economics" which does not constitute real growth (Regan & Brazys, 2021), and thus it is not reasonable to assume that social spending would have an effect. To an extent this could be solved by using GNI instead, however that would bring other challenges. As a result the direct relationship between social spending and GDP per capita is less meaningful in this context in these countries.

With the effect of these two countries lowered, the model captures a more representative picture of continental Europe, where social spending plays a structurally significant role in shaping both economic stability and growth. The consistency of the positive multiplier effect, even after accounting for country-specific fixed effects, provides strong support for the view that fiscally active welfare states contribute positively to economic outcomes - not only through direct government expenditure, but also through their broader effects on consumption and investment behavior.

5. Limitations and Recommendations

While this thesis offers strong empirical evidence for a positive relationship between social spending and GDP per capita across European countries, it is not without limitations. First, the study relies exclusively on GDP per capita as the measure of economic performance. Although this is a widely accepted and consistently reported metric (Barro & Sala-i-Martin, 2004), it does not capture broader aspects of societal well-being such as income distribution, education, health, or environmental sustainability Stiglitz et al. (2009). Future studies should consider integrating multidimensional indicators like the Gini coefficient to reflect inequality, the HDI for a composite view of development, or subjective metrics such as the World Happiness Index which although not perfect might broaden the scope (Carlsen, 2018). These could offer a more holistic understanding of how public expenditure affects well-being beyond aggregate output. Also the fact that GDP is partly calculated using government expenditure, social spending mechanically inflates it, so adjusting the multiplier should be addressed.

A second limitation is the exclusion of dynamic econometric models, such as Arellano-Bond or system GMM, which are commonly used in growth regressions to account for time-lagged effects and endogeneity (Roodman, 2009). This thesis employed a first difference estimator using a robust linear model, which, while robust and appropriate for controlling unobserved heterogeneity, does not capture the delayed economic impact of social spending. Moreover, the analysis was based on annual data, which limits the granularity of the model. Future research could benefit from utilizing quarterly data, where available, to capture shorter-term policy effects and improve model precision. Enhanced temporal resolution could also increase the within-country R-squared, potentially providing a clearer view of cyclical spending effects.

The use of nominal GDP per capita rather than GDP in purchasing power parity (PPP) or GNI presents another limitation. As discussed in the results section, this decision necessitated the exclusion of Ireland, where GDP is inflated by foreign multinationals exploiting favorable tax laws (OECD, 2016) and Norway, where GDP is heavily influenced by returns from the sovereign wealth fund. These countries do not conform to the assumed relationship between public social spending and domestic economic output. Future research using PPP-adjusted GDP or GNI could include these states without distortion, improving both sample representativeness and comparative validity.

Geographically, the scope of the study was confined to European countries, excluding some Eastern European states due to incomplete data and omitting the United Kingdom because of post-Brexit data limitations. While this helped ensure data consistency using Eurostat, it limits the external validity of the findings. As Cooray and Nam (2024) highlight, the effectiveness of social spending is highly context-dependent and may vary across regions with different levels of institutional capacity, governance quality, and fiscal transparency. Future studies could extend this analysis to non-European countries, especially in emerging markets or developing economies, to test whether similar relationships hold and under what institutional or demographic conditions they diverge.

Another significant limitation of this study is the absence of control variables that could account for the influence of other factors affecting economic performance. By focusing solely on the relationship between social spending and GDP per capita, the analysis does not isolate the effect of social expenditure from other potentially impactful variables such as education levels, labor market dynamics, trade openness, technological progress, or institutional quality. The exclusion of these variables limits the ability to draw more precise causal inferences and may result in omitted variable bias, as social spending does not operate in a vacuum but rather within a complex web of economic and social influences.

Finally, this thesis treats social spending as a single aggregated variable, without differentiating between direct transfers (e.g., pensions, unemployment benefits) and indirect expenditures (e.g., tax breaks with social purposes). While this broad measure aligns with OECD and Eurostat classifications (Eurostat, 2023; OECD, 2019), it limits the ability to assess the efficiency or targeting of specific policy instruments. Future research should aim to disaggregate social spending by function, ideally using the COFOG framework, to determine which categories yield the highest returns in terms of growth or well-being. Such granularity would offer a more nuanced test of Keynesian multiplier effects (Musgrave, 1959) and allow for evaluation against neoclassical critiques that warn of inefficiencies and potential crowding-out of private investment. This could lead to more refined policy insights regarding the optimal composition—not just the volume—of government social expenditure.

Conclusion

This thesis set out to explore the relationship between government social spending and economic activity, as measured by GDP per capita, across many European countries over a twenty-year period. Utilizing robust econometric techniques, particularly the first difference estimator, and data sourced from Eurostat, the analysis identified a strong and statistically significant positive association between social spending and GDP per capita. Even after adjusting for structural outliers - namely Ireland and Norway - the relationship remained stable, with each additional euro of social expenditure per capita associated with an approximate 2.94 € increase in GDP per capita in first differences.

These findings provide empirical support for Keynesian economic theory, suggesting that well-managed social spending in institutionally stable environments likely contributes meaningfully to economic output. Social expenditure not only serves redistributive and welfare functions but also operates as a fiscal multiplier by stimulating demand and fostering investment-friendly conditions. The consistency of results across the dataset reinforces the conclusion that social spending plays a valuable role in driving sustainable economic growth within European welfare states.

Nonetheless, the study acknowledges several important limitations. The use of GDP per capita as a singular measure of economic activity, the static modeling framework, and the aggregation of all social spending types limit the granularity and scope of inference. Future research should expand on these areas by incorporating multidimensional well-being indicators, exploring dynamic econometric models, and disaggregating expenditure categories to better understand the efficacy of specific policy tools.

From a policy perspective, these findings underscore the economic utility of maintaining and optimizing social welfare systems - not just as tools of equity and protection, but as engines of growth. For European governments navigating the tension between fiscal responsibility and social commitments, this thesis offers a data-driven foundation for prioritizing smart, targeted public investment that yields both economic and social dividends.

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Appendices

A. Data

The data used for the analysis is available in the attached file or at this link/qr code:

https://docs.google.com/spreadsheets/d/e/2PACX-1vTsOTmwLn9q58-tqtEasVIBwpCE_0apMNCeXdtthMEvQdj5fuvgCf29ZDyXMd7BUjYZnhCV2fnzYjjj/pub?output=xlsx

B. Code

The code used the analysis is available in the attached file or using the link/qr code.

https://colab.research.google.com/drive/1SaqvSTSm2nn8UVub0kq5dkIC2j0e1feN?usp=sharing